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22IT301/22CS301/22IS301/22CT301/22CD301 - PROBABILITY, STATISTICS AND QUEUING THEORY

UNIT IV - DESIGN OF EXPERIMENTS AND CONTROL CHARTS

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1. CONTROL CHART FOR MEARURMENTS (X AND R CHARTS)

1.1 Quality Control

Quality Control is a systematic control of various factors that affect the quality of the product. The various factors include material, tools, machines, type of labour, working conditions, measuring instruments, etc. Quality Control can be defined as the entire collection of activities which ensures that the operation will produce the optimum Quality products at minimum cost.

- To decide about the standard of quality of a product that is easily acceptable to the customer and at the same time this standard should be economical to maintain.
- To take different measures to improve the standard of quality of product.
- To take various steps to solve any kind of deviations in the quality of the product during manufacturing.

1.2 Advantages of Quality Control

- Quality of product is improved which in turn increases sales.
- Scrap rejection and rework are minimized thus reducing wastage. So the cost of manufacturing reduces.
- Good quality product improves reputation.
- Inspection cost reduces to a great extent.
- Uniformity in quality can be achieved.
- Improvement in manufacturer and consumer relations.

1.3 Statistical Quality Control (S.Q.C):

Statistics means data, a good amount of data to obtain reliable results. The science of statistics handles this data in order to draw certain conclusions.

This is a quality control system employing the statistical techniques to control quality by performing inspection, testing and analysis to conclude whether the quality of the product is as

per the laid quality standards.

Using statistical techniques, S.Q.C. collects and analyses data in assessing and controlling product quality.

1.4 Process Control

Under this the quality of the products is controlled while the products are in the process of production.

Control charts are also used in the field of advertising, packing etc. They ensure that whether the products confirm to the specified quality standard or not.

Process Control consists of the systems and tools used to ensure that processes are well defined, performed correctly, and maintained so that the completed product conforms to established requirements. Process Control is an essential element of managing risk to ensure the safety and reliability of the Space Shuttle Program.

2. CONTROL CHARTS

Since variations in manufacturing process are unavoidable, the control chart tells when to leave a process alone and thus prevent unnecessary frequent adjustments. Control charts are graphical representation and are based on statistical sampling theory, according to which an adequate sized random sample is drawn from each lot. Control charts detect variations in the processing and warn if there is any departure from the specified tolerance limits. These control charts immediately tell the undesired variations and help in detecting the cause and its removal. In control charts, where both upper and lower values are specified for a quality characteristic, as soon as some products show variation outside the tolerances, a review of situation is taken and corrective step is immediately taken.

If analysis of the control chart indicates that the process is currently under control (i.e. is stable, with variation only coming from sources common to the process) then data from the process can be used to predict the future performance of the process. If the chart indicates that the process being monitored is not in control, analysis of the chart can help determine the sources of variation, which can then be eliminated to bring the process back into control. A control chart is a specific kind of run chart that allows significant change to be differentiated from the natural

variability of the process.

Control chart is:

• A device which specifies the state of statistical control,

• A device for attaining statistical control,

• A device to judge whether statistical control has been attained or not.

1. A control charts indicates whether the process is in control or out of control. 2. It determines process variability and detects unusual variations taking place in a process. 3. It ensures product quality level.

4. It warns in time, and if the process is rectified at that time, scrap or percentage rejection can be reduced.

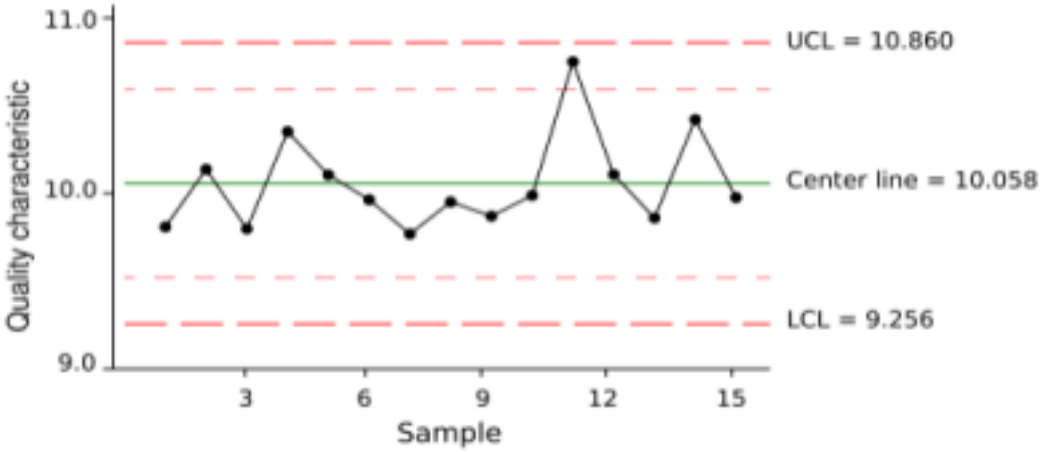
5. It provides information about the selection of process and setting of tolerance limits. 6. Control charts build up the reputation of the organization through customer's satisfaction.

• Points representing a statistic (e.g., a mean, range, proportion) of measurements of a quality characteristic in samples taken from the process at different times [the data] • The mean of this statistic using all the samples is calculated (e.g., the mean of the means, mean of the ranges, mean of the proportions)

• A center line is drawn at the value of the mean of the statistic

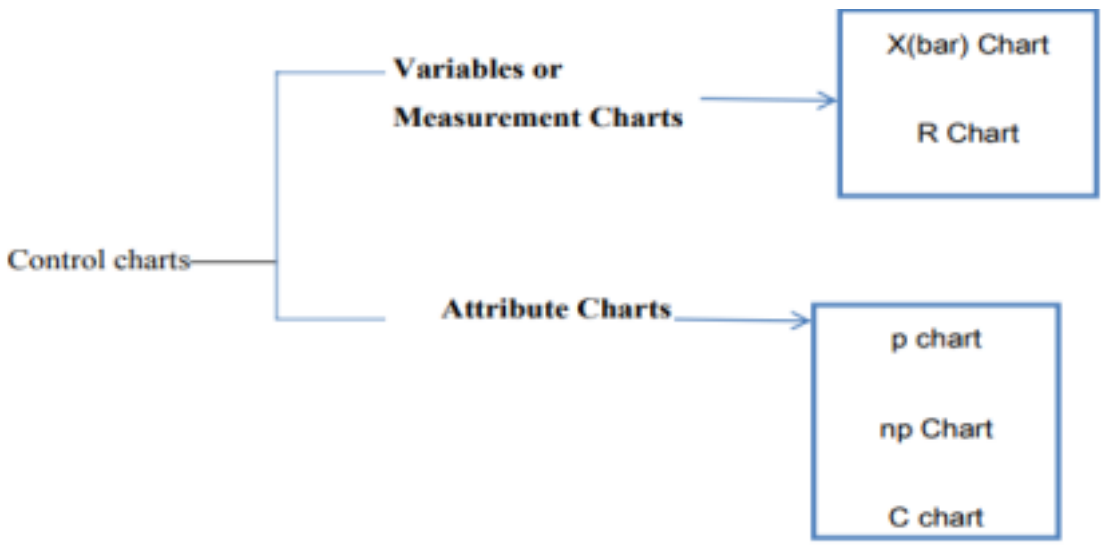
• The standard error (e.g., standard deviation/sqrt(n) for the mean) of the statistic is also calculated using all the samples

• Upper and lower control limits (sometimes called "natural process limits") that indicate the threshold at which the process output is considered statistically 'unlikely' are drawn typically at 3 standard errors from the center line



2.1 Types of Control Charts

Control charts can be used to measure any characteristic of a product, such as the weight of a cereal box, the number of chocolates in a box, or the volume of bottled water. The different characteristics that can be measured by control charts can be divided into two groups: **variables and attributes.**

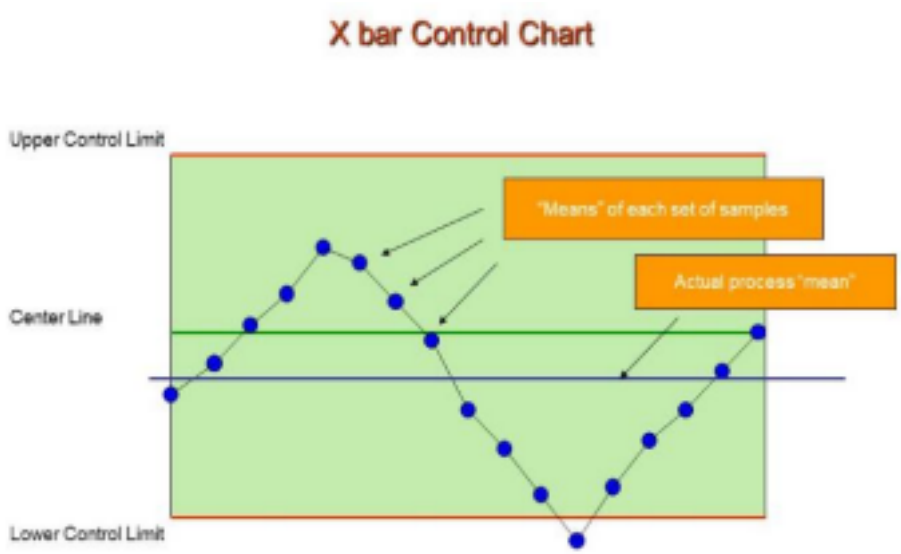


- **A control chart for variables** is used to monitor characteristics that can be measured and have a continuum of values, such as height, weight, or volume. A soft drink bottling operation is an example of a variable measure, since the amount of liquid in the bottles is measured and can take on a number of different values. Other examples are the weight of a bag of sugar, the temperature of a baking oven, or the diameter of plastic tubing.
- **A control chart for attributes**, on the other hand, is used to monitor characteristics that have discrete values and can be counted. Often they can be evaluated with a simple yes or no decision. Examples include color, taste, or smell. **The monitoring of attributes**

usually takes less time than that of variables because **a variable needs to be measured** (e.g., the bottle of soft drink contains 15.9 ounces of liquid). **An attribute requires only a single decision**, such as **yes or no, good or bad, acceptable or unacceptable** (e.g., the apple is good or rotten, the meat is good or stale, the shoes have a defect or do not have a defect, the light bulb works or it does not work) or counting the number of defects (e.g., the number of broken cookies in the box, the number of dents in the car, the number of barnacles on the bottom of a boat).

2. 2 X - Bar Control Chart:

This type of chart graphs the means (or averages) of a set of samples, plotted in order to monitor the mean of a variable, for example the length of steel rods, the weight of bags of compound, the intensity of laser beams, etc.. In constructing this chart, samples of process outputs are taken at regular intervals; the means of each set of samples are calculated and graphed onto the X bar control chart. This chart can then be utilized to determine the actual process mean, versus a nominal process mean and will demonstrate if the mean output of the process is changing over time.



2.3 Range "R" Control Chart

This type of chart demonstrates the variability within a process. It is suited to processes

where the sample sizes are relatively small, for example <10. Sets of sample data are recorded from a process for the particular quality characteristic being monitored. For each set of date the difference between the smallest and largest readings are recorded.

This is the range "R" of the set of data. The ranges are now recorded onto a control chart. The center line is the averages of all the ranges.

When the sample mean and standard deviation is given,

$$\text{X bar chart}: UCL_{\bar{x}} = \bar{\bar{x}} + 3\frac{\hat{\sigma}}{\sqrt{n}}$$

$$LCL_{\bar{x}} = \bar{\bar{x}} - 3\frac{\hat{\sigma}}{\sqrt{n}}.$$

	Central line		Control limits	
	Definition	Formula	Upper limit	Lower limit
Xbar-chart	Average on *m* samples of the quality characteristic averages over *n* units tested per sample	$\bar{\bar{X}} = \frac{\sum_{j=1}^{m} \bar{X}_j}{m}$	$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \bar{R}$	$LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \bar{R}$
R-chart	Average on *m* samples of the ranges calculated for each sample	$\bar{R} = \frac{\sum_{j=1}^{m} R_j}{m}$	$UCL_R = D_4 \bar{R}$	$LCL_R = D_3 \bar{R}$

3. EXAMPLES

3.1 Example:

A machine drills hole in a pipe with a mean diameter of 0.532 cm and a standard deviation of 0.002 cm. Calculate the control limits for mean of samples 5.

Solution:

Given $\bar{\bar{X}} = 0.532$, $\sigma = 0.002$, $n = 5$

The control limits for $\bar{X}$ chart is

$$UCL = \bar{\bar{X}} + 3\frac{\sigma}{\sqrt{n}} = 0.532 + 3\frac{0.002}{\sqrt{5}} = 0.5346$$

$$CL = \bar{\bar{X}} = 0.532$$

$$UCL = \bar{\bar{X}} - 3\frac{\sigma}{\sqrt{n}} = 0.532 - 3\frac{0.002}{\sqrt{5}} = 0.5293$$

3.2 Example

The following data gives the readings for 8 samples of size 6 each in the production of a certain product. Find the control limits using mean chart.

Sample	1	2	3	4	5	6
Mean	300	342	351	319	326	333
Range	25	37	20	28	30	22

Given for n = 6, $A_2 = 0.483$.

Solution:

Sample	1	2	3	4	5	6	**Total**
Mean	300	342	351	319	326	333	**1971**
Range	25	37	20	28	30	22	**162**

Table 9.20

The control limits for $\bar{X}$ chart is

$$\bar{\bar{X}} = \frac{\sum \bar{X}}{number\ of\ samples} = \frac{1971}{6} = 328.5 \qquad \bar{R} = \frac{\sum R}{n} = \frac{162}{6} = 27$$

$$UCL = \bar{\bar{X}} + A_2\bar{R} = 328.5 + 0.483(27) = 341.54$$

$$CL = \bar{\bar{X}} = 328.5$$

$$LCL = \bar{\bar{X}} - A_2\bar{R} = 328.5 - 0.483(27) = 315.45$$

3.3 Example

The data shows the sample mean and range for 10 samples for size 5 each. Find the control limits for mean chart and range chart.

Sample	1	2	3	4	5	6	7	8	9	10
Mean	21	26	23	18	19	15	14	20	16	10
Range	5	6	9	7	4	6	8	9	4	7

Solution:

Sample	1	2	3	4	5	6	7	8	9	10	**Total**
Mean	21	26	23	18	19	15	14	20	16	10	**182**
Range	5	6	9	7	4	6	8	9	4	7	**65**

The control limits for $\overline{X}$ chart is

$$\overline{\overline{X}} = \frac{\sum \overline{X}}{number\ of\ samples} = \frac{182}{10} = 18.2 \qquad \overline{R} = \frac{\sum R}{n} = \frac{65}{10} = 6.5$$

$$UCL = \overline{\overline{X}} + A_2\overline{R} = 18.2 + 0.577(6.5) = 21.95$$

$$CL = \overline{\overline{X}} = 18.2$$

$$LCL = \overline{\overline{X}} - A_2\overline{R} = 18.2 - 0.577(6.5) = 14.5795$$

The control limits for Range chart is

$$UCL = D_4\overline{R} = 2.114(6.5) = 13.741$$

$$CL = \overline{R} = 6.5$$

$$LCL = D_3\overline{R} = 0(6.5) = 0$$

3.4 Example

The following data gives readings of 10 samples of size 6 each in the production of a certain product. Draw control chart for mean and range with its control limits.

Sample	1	2	3	4	5	6	7	8	9	10
Mean	383	508	505	582	557	337	514	614	707	753
Range	95	128	100	91	68	65	148	28	37	80

Solution:

Sample	1	2	3	4	5	6	7	8	9	10	**Total**
Mean	383	508	505	582	557	337	514	614	707	753	**5460**
Range	95	128	100	91	68	65	148	28	37	80	**840**

$$\bar{\bar{X}} = \frac{\sum \bar{X}}{10} = \frac{5460}{10} = 546 \qquad \bar{R} = \frac{\sum R}{n} = \frac{840}{10} = 84$$

$$UCL = \bar{\bar{X}} + A_2\bar{R} = 546 + 0.483(84) = 586.57$$

$$CL = \bar{\bar{X}} = 546$$

$$LCL = \bar{\bar{X}} - A_2\bar{R} = 546 - 0.483(84) = 505.43$$

The control limits for Range chart is

$$UCL = D_4\bar{R} = 2.004(84) = 168.336$$

$$CL = \bar{R} = 84$$

$$LCL = D_3\bar{R} = 0(84) = 0$$

3.5 Example

You are given below the values of sample mean (X) and the range (R) for ten samples of size 5 each. Draw mean chart and comment on the state of control of the process.

Sample number	1	2	3	4	5	6	7	8	9	10
$\bar{X}$	43	49	37	44	45	37	51	46	43	47
R	5	6	5	7	7	4	8	6	4	6

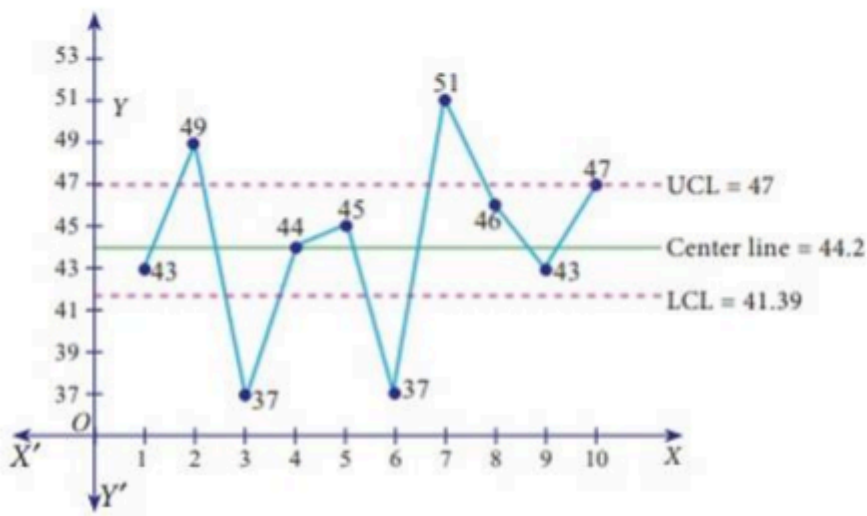
Given the following control chart constraint for: n = 5, $A_2 = 0.58$, $D_3 = 0$ and $D_4 = 2.115$

Solution:

$$\bar{\bar{X}} = \frac{\sum \bar{X}}{10} = \frac{442}{10} = 44.2$$

$$\bar{R} = \frac{\sum R}{n} = \frac{58}{10} = 5.8$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$
$$= 44.2 + 0.483(5.8) = 47.00$$
$$CL = \bar{\bar{X}} = 44.2$$
$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 44.2 - 0.483(5.8) = 41.39$$



The above diagram shows all the three control lines with the data points plotted, since four points falls out of the control limits, we can say that the process is out of control.

DISCUSSION QUESTIONS:

1, How to find mean of p, c and np

2. Define Lower and upper control limits

3. What do you mean by stability of the system